Eggshell–Agar Biocomposites

Development and characterisation for sustainable interior applications

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The waste



94%
calcium carbonate

> 12
months of collection

1
local bakery

Porous, protein-bound crystalline structure. Specific gravity 0.85–2.66 — lighter than most mineral fillers.

The problem

ESTABLISHED PRACTICE

Mineral phase

waste eggshell, 5–30% of matrix

Binding phase

Portland clinker · bitumen · lime · ceramic body · PLA, epoxy, PVC, PU

The waste is valorised. The matrix is not.

THIS STUDY

Mineral phase

waste eggshell, 66% of solids

Binding phase

agar, plasticised with glycerol

Both phases biological and biodegradable.

Teara et al. 2019 · Adogla et al. 2016 · Razzaq & Yousif 2017 · Soundara & Vilasini 2015 · Vilarinho et al. 2022 · Owuamanam & Cree 2020

The binder

Agar is a thermoreversible polysaccharide from red seaweed. Its behaviour with glycerol has been mapped systematically — 32 formulations, released as an open dataset.

Hernández et al. (2022), Materials 15(11), 3954

BUT ONLY AS A FILM

Unfilled, thin, tested in uniaxial tension, for packaging. Never as the binder of a mineral-filled cast solid in bending.

THE SHIFT

—	Film literature	This study
Form	Thin cast film	Cast solid, 20–50 mm
Filler	Unfilled	Mineral, 66% of solids
Loading	Uniaxial tension	Compression, bending
Use	Packaging	Interior surfaces

The compositional logic carries over. The form and the loading case do not.

The gap

Materiom publishes biomaterial recipes under CC-BY-SA, archives each as prior art, and indexes property data with citable DOIs. The infrastructure for shared procedures is built.

4

recipes with published property data

WHICH ARE

agar bioplastic
gelatin bioplastic
mycelium–walnut shell
mussel shell–sucrose ceramic

WHAT THAT MEANS

The nearest analogue is the mussel shell–sucrose ceramic: a shell mineral in a biological binder. Nothing couples a shell mineral to agar. Nothing reports compressive, flexural, ignitability and hygric data for one formulation family.

A protocol and a property set entering that commons.

materiom.org/data, accessed October 2026

Aim and scope

AIM

To document a reproducible production protocol for a fully bio-based eggshell composite, and to characterise what it can and cannot do.

NOT IN SCOPE

- Structural or loadbearing use
- Durability and accelerated ageing
- Reaction-to-fire classification
- Life-cycle assessment

IN SCOPE

Dimension	This study
Formulations	Five — three compression, two flexure
Specimens	Three per formulation
Properties	Compression, flexure, ignitability, absorption, density
Standards	ASTM D790-17 · ISO 11925-2 · ASTM D570
Conditions	Ambient cure, single laboratory
Application	Non-loadbearing interior surfaces

Method

1

Produce

Develop the composite and record every parameter

Iterative formulation, over one year

2

Measure

Characterise against published standards

Compression · flexure · fire · absorption

3

Position

Benchmark against two material families

Conventional interior · bio-based composites

4

Propose

Derive an evaluation criterion from the behaviour

Deformation capacity alongside strength

Material-driven: the properties are measured first and the application is derived from them, rather than the material being selected to fit a predetermined form. The design proposal shown later follows from the measurements, not the other way round.

Questions

01 Which formulation variable governs compressive strength?

02 What does plasticiser content do to flexural behaviour?

03 How does it compare with conventional interior materials?

04 How does it compare with other bio-based composites?

05 Is compressive strength the right criterion at all?

HYPOTHESIS

A fully bio-based eggshell–agar composite can be produced repeatably under ambient conditions, and its performance for interior surfaces is governed by deformation capacity rather than by strength.



Four constituents, all biodegradable

Waste eggshell powder · agar · glycerol · water, with a shellac finish. Colour and texture vary from natural pigments and mould geometry at constant composition.

Stage 1 — the mineral



Whole shell	Cleaned	Heat-treated	Milled	Sieved
collected from one bakery	membrane and residue removed by hand	decontamination and drying	laboratory mechanical grinder	passing fraction only
—	—	**120 °C, 20 min**	**to homogeneous powder**	**150 µm mesh**

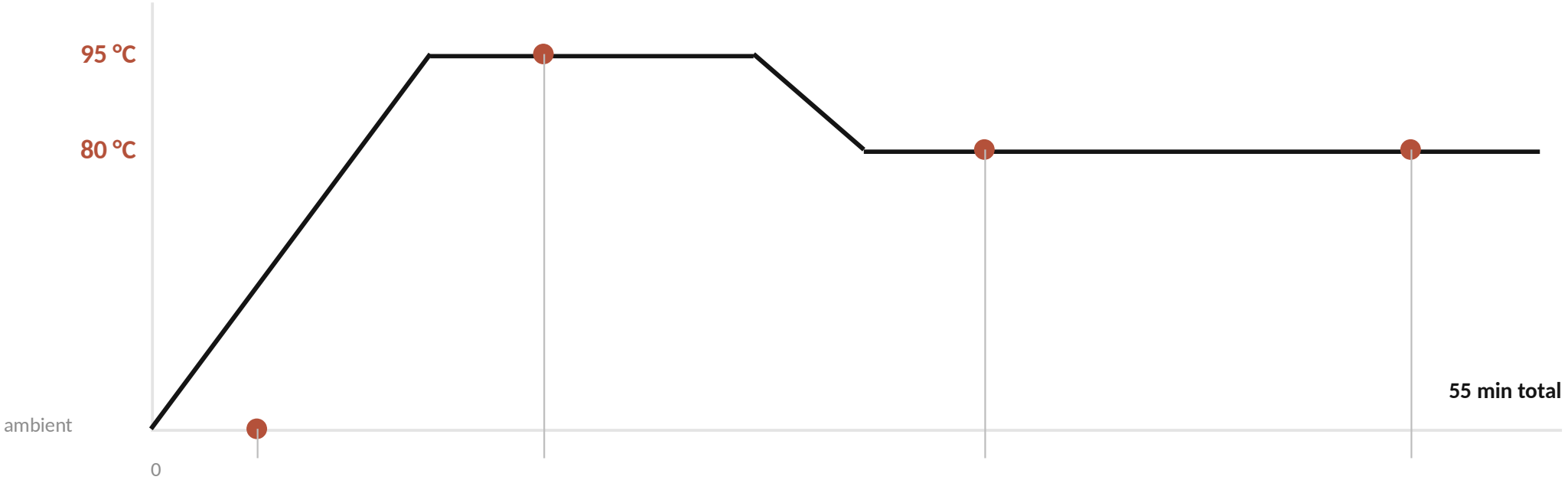
WHY 150 MM

Chosen to match the particle size used in the eggshell polymer-composite literature, so the mineral phase here is comparable with that work.

WHY NOT STERILISATION

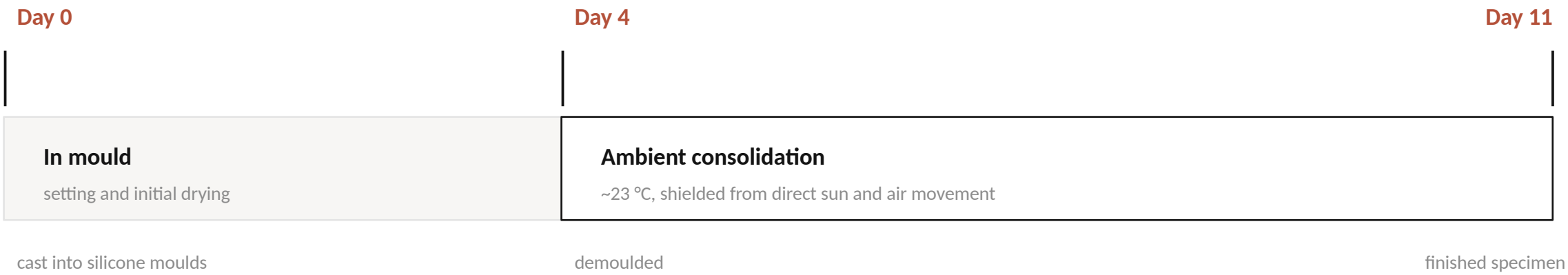
Dry-heat sterilisation needs far higher temperatures for far longer, and sterility was not verified. This step is decontamination and drying.

Stage 2 — the binder



Schematic. The two set points and the total heating time are recorded; intermediate ramp rates are not, and are drawn indicatively. Stirring was manual throughout: shear rate is not a controlled quantity.

Stage 3 — casting and cure



WHY SILICONE

Flexible enough to release without a parting agent and without damaging the partially consolidated specimen. 50 mm cube moulds for compression, flat moulds otherwise.

NO OVEN AT ANY STAGE

After the 55-minute cook the material receives no further heat input. The entire cure is at room temperature.

Disclosure the composite shrinks appreciably as water evaporates. Cross-sections were measured at test time, but post-drying dimensions were not logged, so every dimension quoted in this talk is a mould dimension.

Formulations

ID	Test	Water g	Shell g	Agar g	Glycerol g	Solids g	Shell : binder	Solids : water	Glycerol % solids
C1	Compression	325	80	35	5	120	**2.00**	0.37	4.2
C2	Compression	310	90	40	5	135	**2.00**	0.44	3.7
C3	Compression	280	110	45	10	165	**2.00**	0.59	6.1
F1	Flexure	450	190	70	60	320	1.46	0.71	18.8
F2	Flexure	480	190	70	30	290	1.90	0.60	10.3

COMPRESSION SERIES

Shell : binder is held at exactly 2.00. The only variable is solids concentration.

FLEXURAL SERIES — CONFOUND

Glycerol was halved at fixed batch mass, so water rose with it. The plasticiser effect is not isolated.

Three specimens per formulation. Values are means; individual specimen values were not retained, so standard deviations are not available.

Tests

Property	Standard	Specimen	n	Measured
Compressive behaviour	Cube compression	50 mm cube moulds	3 per mix	Max load; compressive stress
Flexural behaviour	ASTM D790-17	180 × 60 × 20 mm, 120 mm span	3 per mix	Flexural strength; modulus
Reaction to fire	ISO 11925-2:2020	40 × 50 mm, 45°, 15 s edge flame	2	Time to ignition; to extinguish
Water absorption	ASTM D570-98	40 × 50 mm, shellac-coated	1	Mass gain, 24 h immersion
Bulk density	Gravimetric	50 mm cube	1	Mass / mould volume

Mechanical values are means of three specimens; individual values were not retained, so standard deviations cannot be reported. Fire, absorption and density rest on single determinations and are indicative, not characteristic.

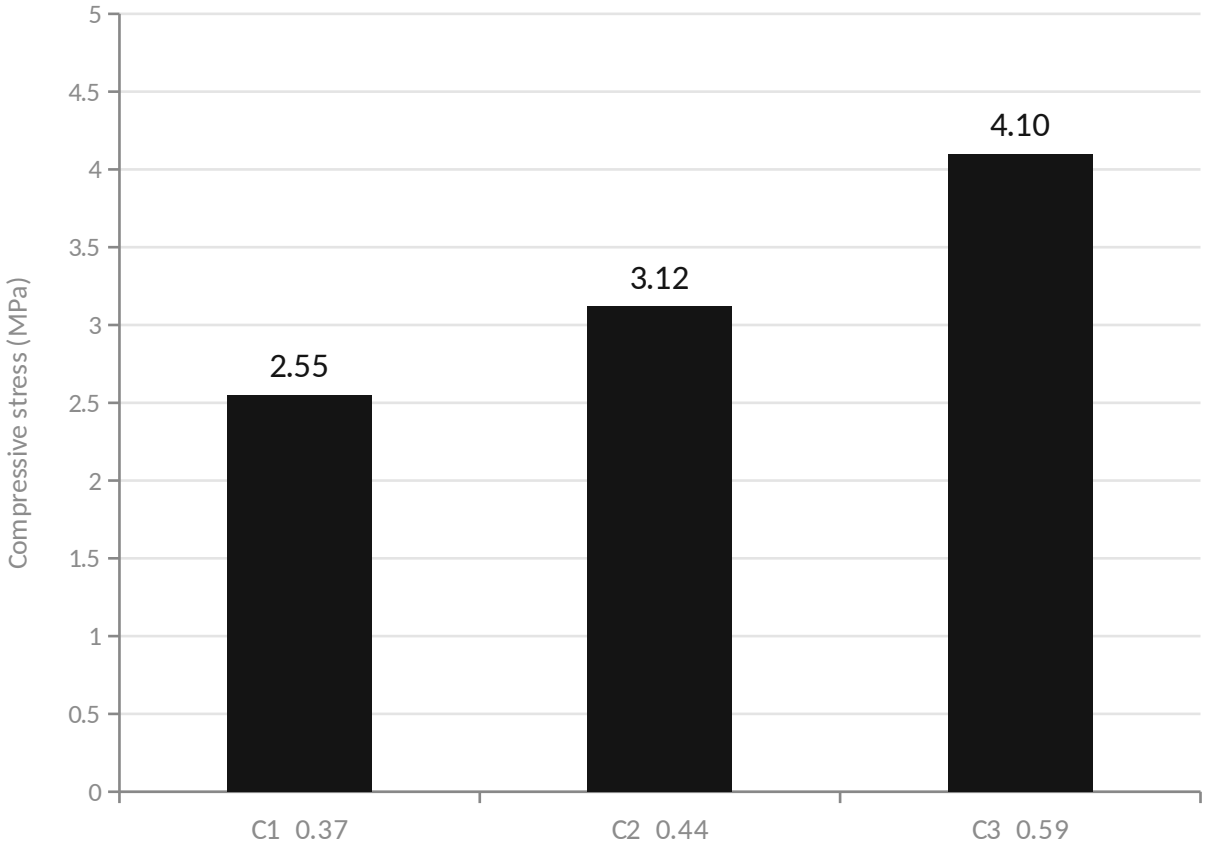
All testing: Advanced Construction Materials Research Laboratory, Hacettepe University.

Compression

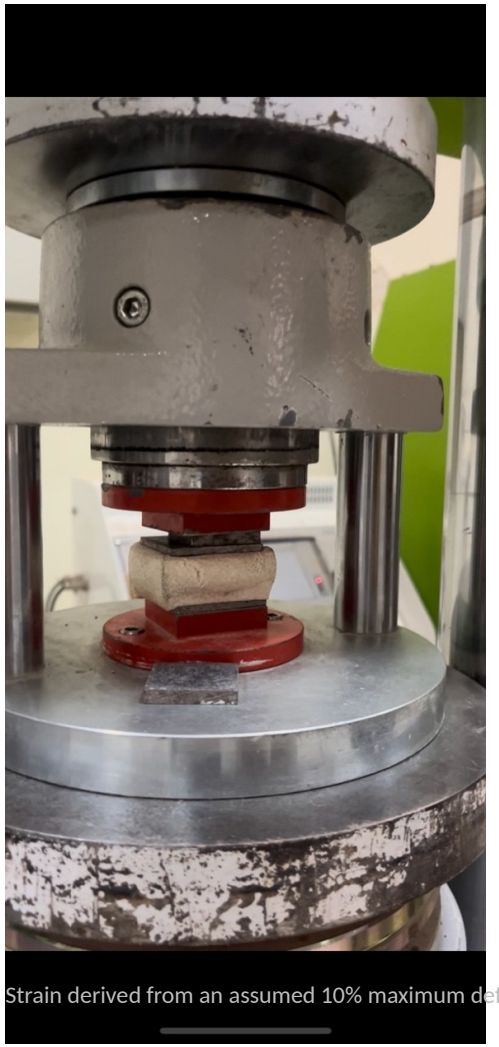
ID	Solids : water	F_{max} kN	σc MPa
C1	0.37	3.89	2.55
C2	0.44	4.30	3.12
C3	0.59	**6.80**	**4.10**

Strength rises with solids concentration, not with mineral content.

Shell : binder is fixed at 2.00 across the series, so packing density is the only variable. Linear across the range tested, $R^2 = 0.99$ — three formulations, so a hypothesis, not a law.

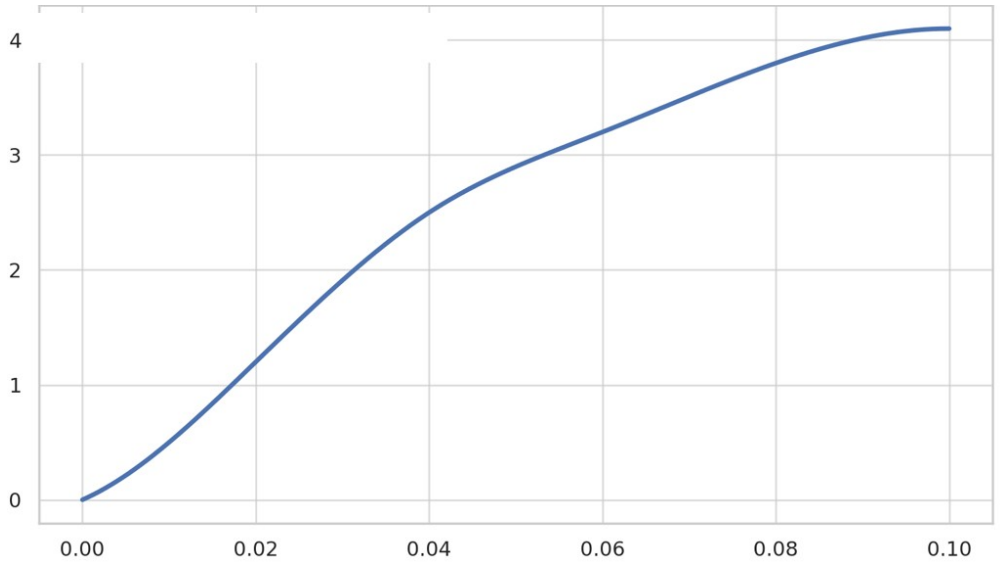


Crushing



Strain derived from an assumed 10% maximum deformation based on observation, not instrumented measurement. · Bulk density 1.38 g/cm³, mould-volume basis.

COMPRESSIVE STRESS – STRAIN, C3



THE CURVE

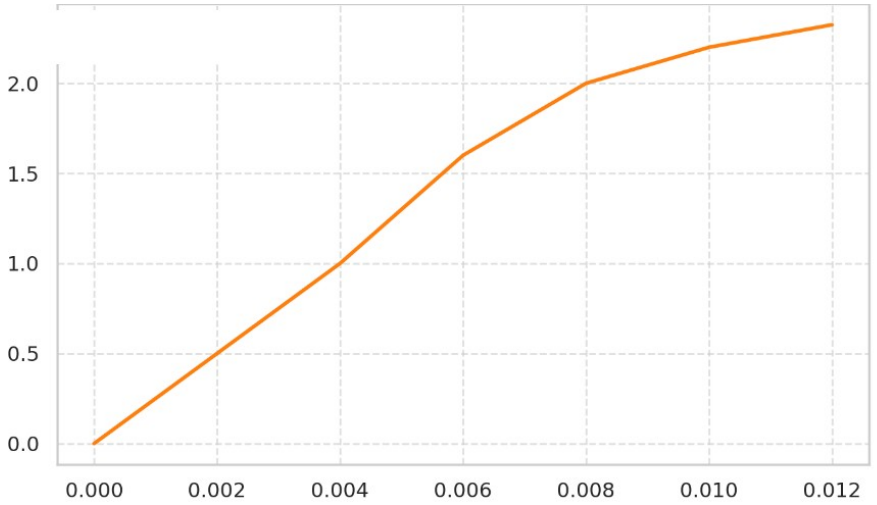
- Linear to about 1 MPa
- Progressive stiffness loss
- Peak near 10% strain
- No abrupt load drop
- Deformation with retained integrity

It crushes.
It does not fracture.

Flexure

ID	Glycerol % solids	F_{max} N	σF MPa	EF MPa	ε peak
F1	18.8	—	*not measurable*	—	—
F2	10.3	310	**2.325**	**192**	0.0121

FLEXURAL STRESS – STRAIN, F2



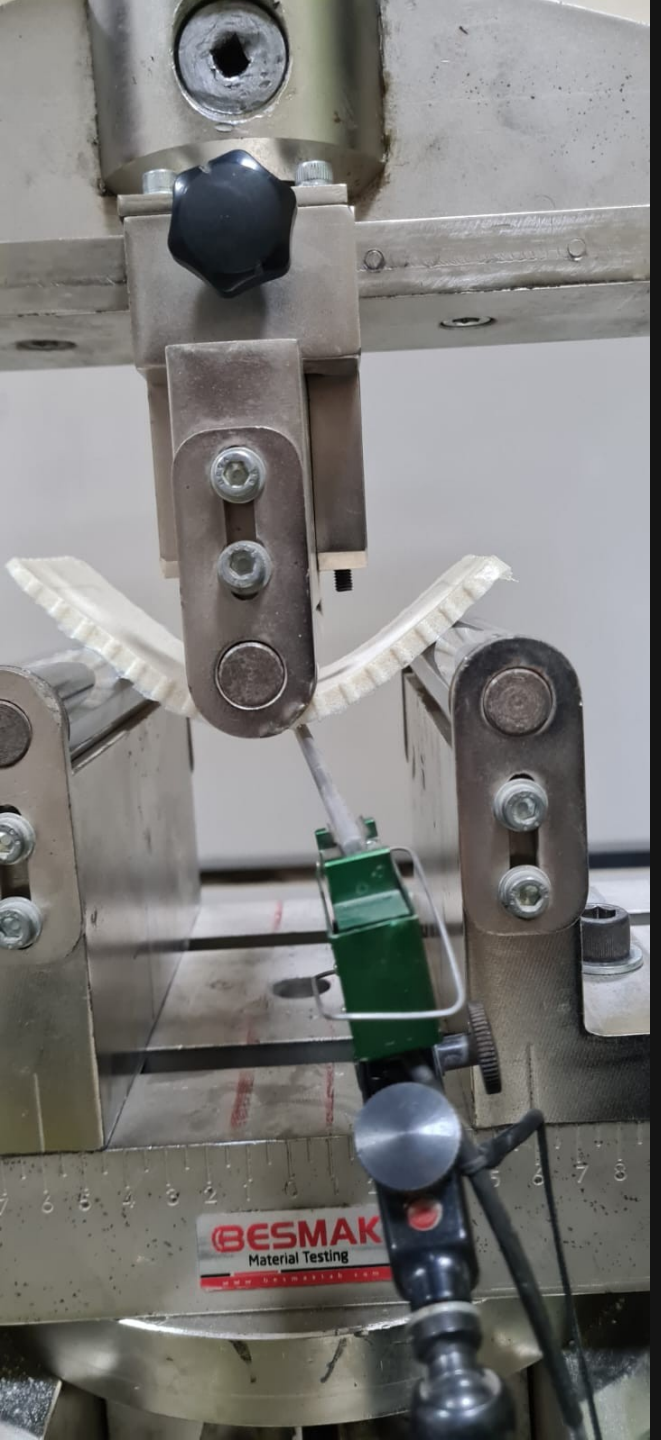
Three specimens per formulation. The two formulations differ in glycerol, but batch mass was fixed, so solids concentration moved as well.

DISCLOSURE

The specimen is kerfed on its tension face. The standard assumes a solid rectangle, and the calculation uses the full section. Part of the compliance is geometric: the modulus is a lower bound.

Plasticiser decides whether it resists at all.

At 18.8% glycerol the machine registered nothing usable. Halving it gave a section that resists while still flexing.







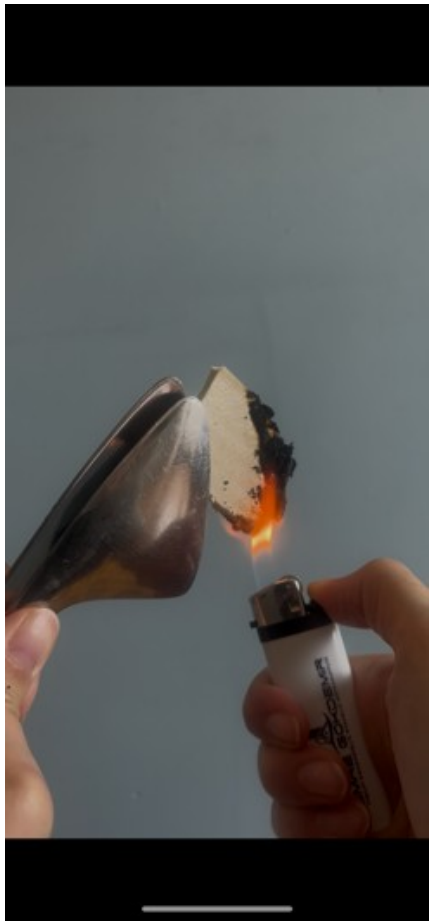
BENDING

The kerfs are visible, and so is the consequence.

Transverse kerfs on the tension face let the panel arch by hand. They also mean the measured compliance is partly geometric, which is why the reported modulus is a lower bound rather than a material constant.

Three-point rig at high deflection · close-up of the kerfed tension face · panel arched by hand

Fire and water



IGNITABILITY — ISO 11925-2

~10 s

to ignition under direct flame

4 s

to self-extinguish

- Charring stays local
- No lateral spread
- No melting, no dripping

Ignitability only. No Euroclass, cone calorimetry, smoke or toxicity data.

ABSORPTION — 24 H IMMERSION

0.7%

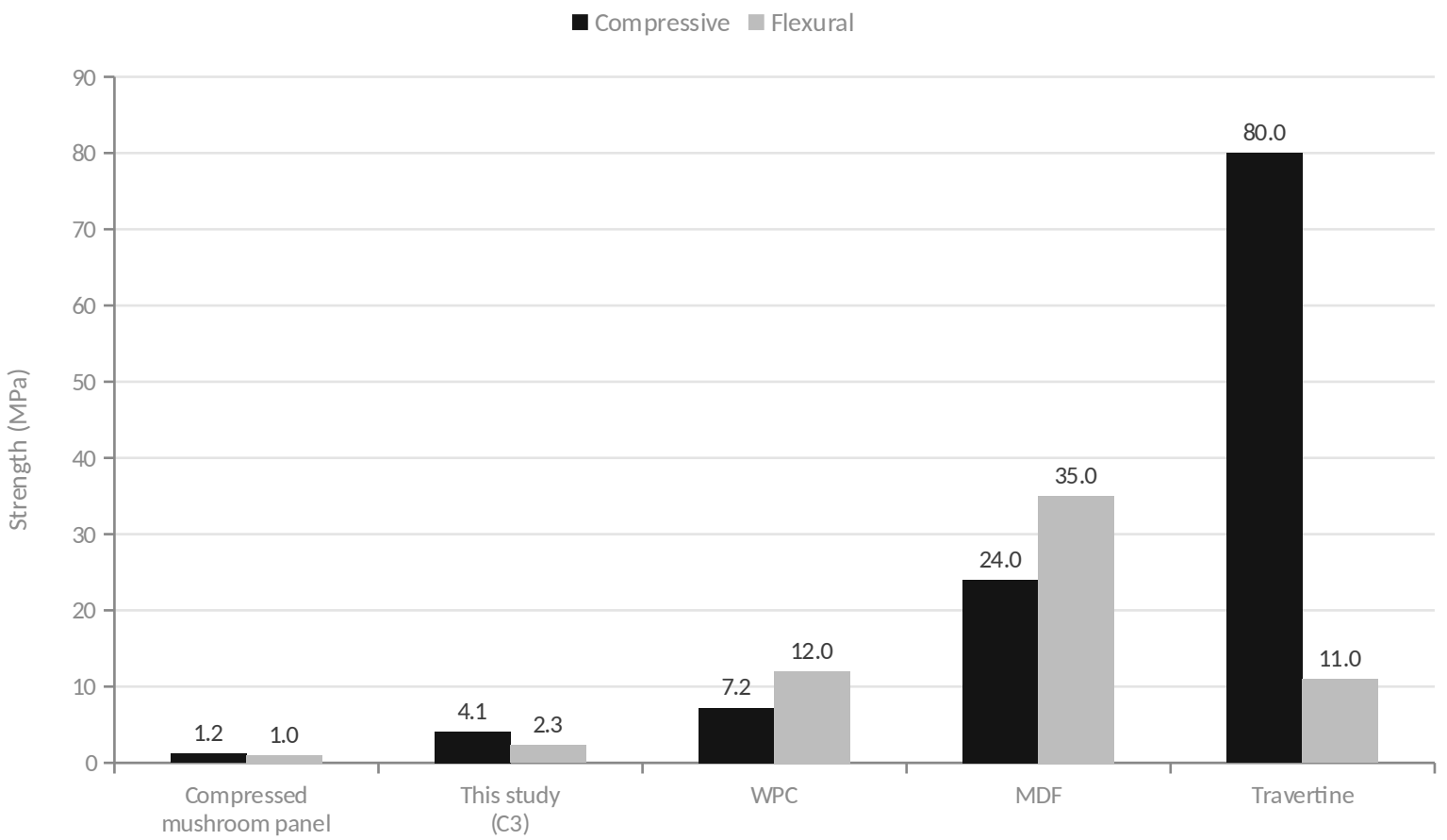
mass gain, shellac-coated. Coating integrity, not the composite, is the limiting factor for humid use.

BULK DENSITY

Material	g/cm³	Reference
This study (C3)	**1.38**	—
Mycelium, cellulose	0.3 – 0.8	Jones et al. 2020
Fired clay brick	1.6 – 2.0	William & Owen 1995
Normal-weight concrete	2.4	William & Owen 1995

Denser than the fungal and cellulosic family, lighter than fired clay. Mould-volume basis, therefore a lower bound.

Versus stone and board



Velasco & Gaona 2011 · Klyosov 2007 · Nemli & Aydın 2007 · Altun 2006

On strength alone it does not compete.

One sixth the compressive strength of MDF. One twentieth that of travertine — a stone it visually resembles. Three times compressed mushroom panel, the only other fully bio-based board here.

Versus bio-based

COMPRESSIVE STRENGTH

Material	σc MPa	Reference
This study (C3)	**4.1**	—
Mycelium composite	0.145 – 0.452	Jones et al. 2020
Hempcrete	0.3 – 0.6	Arnaud & Gourlay 2012
Eggshell-filled PLA	2 – 5	Kim & Pal 2010
Mineral-filled biocomposite	2.1 – 39.6	Shah et al. 2016

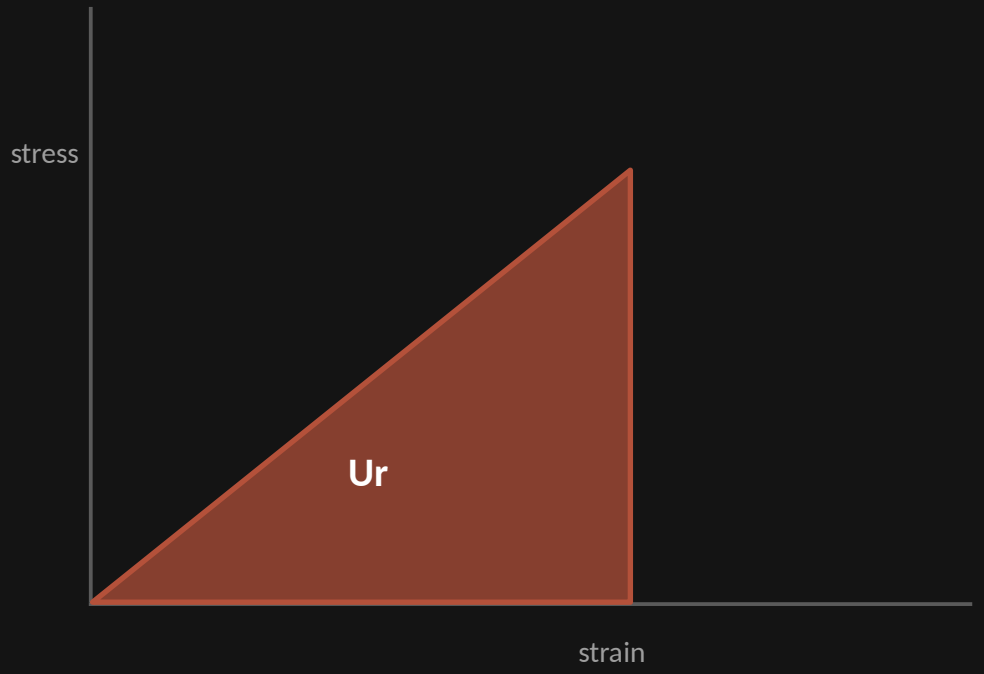
ELASTIC MODULUS

Material	E MPa	Reference
This study (F2)	**192**	—
Mycelium composite	1.8 – 5	Islam et al. 2018
Soy-based bioplastic	124	Mansour et al. 2021
Flax-fibre PLA	198	Faruk et al. 2012
LDPE	100 – 300	Zhao et al. 2022
Wood (pine)	11 000	Pope & Hudson 1989

Water-cast and ambient-dried, at melt-processed stiffness.

An order of magnitude above the fungal and lime-binder family in compression, and inside the bio-polymer band in stiffness — between soy bioplastic and flax-fibre PLA, within the LDPE range. No oven, no autoclave, no extrusion. Wood remains two orders of magnitude above.

Resilience



The shaded area is the elastic energy absorbed per unit volume before permanent damage.

$$Ur = \sigma^2 / 2E$$

14.1 kJ/m³

for the formulation measured here

Stiffness sits in the denominator.

The low modulus that counted against the material on the previous slide is exactly what raises this quantity.

THE LIMIT

A comparative table cannot yet be assembled: the literature reports strength and modulus in separate studies on different formulations. This is a data point and a measurement convention, not a ranking.

Where it fits



Modular interior proposal derived from the measured properties

Non-loadbearing surfaces

self-weight and handling loads only

Dry interiors

not wet rooms, not facades

Away from ignition sources

self-extinguishing, but unclassified

Modular, not sheet

failure replaced one unit at a time

AND THE LOOP IT CLOSES

Bakery waste → Shell powder → One 55-min cook → Cast module → Reversible assembly → Biological cycle

Limits

Measurement

- No dispersion data — means of three, individual values not retained
- Compressive strain assumed from observation, not instrumented
- Shrinkage unquantified — density is a lower bound

Process

- Flexural series confounded — glycerol and solids moved together
- Kerfed section calculated as solid — modulus is a lower bound
- Manual stirring; one laboratory, one operator

Untested

- Durability — humidity, thermal cycling, UV, microbial, ageing
- Reaction-to-fire classification
- Life-cycle assessment

None of these is unknown. Each one is the next experiment.

Next

01	**Repeat with dispersion**	Three or more specimens per formulation with individual values retained, so standard deviations can be reported.
02	**Record post-drying geometry**	Measure and log the dried cross-section, giving a true stress and a true density rather than lower bounds.
03	**Instrument the strain**	Replace the assumed 10% deformation with measured strain, giving a real stress–strain curve and a defensible modulus of resilience.
04	**Test an un-kerfed section**	Separate the geometric contribution from the material one, so the flexural modulus becomes a material constant.

Beyond these four: humidity and thermal cycling, microbial resistance, a reaction-to-fire classification, and a life-cycle assessment to support the environmental argument quantitatively.

Conclusions

01	**A protocol others can repeat**	every parameter reported
02	**Packing density governs strength**	not mineral content
03	**Plasticiser sets brittle or ductile**	the transition is sharp
04	**Bio-polymer stiffness, oven-free**	water-cast, ambient-dried
05	**Strength is not the only criterion**	report σ and E together

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Testing: Advanced Construction Materials Research Laboratory, Hacettepe University, with thanks to Dr Hüseyin İlcan. Derived from the author's doctoral dissertation, Hacettepe University, 2025.

References

REVIEWS AND STATE OF THE ART

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- Owuamanam, S. & Cree, D. (2020). Progress of bio-calcium carbonate waste eggshell and seashell fillers in polymer composites. Journal of Composites Science, 4(2), 70.
- Biogenic calcium carbonate from waste shells for advanced material applications (2022). Frontiers in Materials, 9, 1024977.

AGAR AND BIO-POLYMER SYSTEMS

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- Materiom (2026). Open biomaterial recipe commons and property database. materiom.org/data.
- Mansour et al. (2021) · Faruk et al. (2012) · Zhao et al. (2022) — bio-polymer elastic moduli.

COMPARATIVE MATERIAL DATA

- Jones, M. et al. (2020) · Islam et al. (2018) — mycelium composites. Arnaud & Gourlay (2012) — hempcrete.
- Kim & Pal (2010) — eggshell-filled PLA. Shah et al. (2016) — mineral-filled biocomposites.
- Nemli & Aydın (2007) — MDF. Altun (2006) — travertine. Klyosov (2007) — WPC. Velasco & Gaona (2011) — mushroom panel.

Standards: ASTM D790-17 · ISO 11925-2:2020 · ASTM D570-98(2018).